EXERCISE 10

Trajectories of an Uncoupled System by the Chain Rule

Original statement · probls_ch4-i_2627_v2.pdf, p. 3

Chain Rule • Separable Equation • Orbit Equation

STATEMENT

Using the chain rule for $y = y(x(t))$,

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} \implies \frac{dy}{dx} = \frac{\dfrac{dy}{dt}}{\dfrac{dx}{dt}}.$$

Therefore, if

$$\frac{dx}{dt} = \lambda_1 x, \qquad \frac{dy}{dt} = \lambda_2 y$$

we have

$$\frac{dy}{dx} = \frac{\lambda_2 y}{\lambda_1 x}.$$

Solve the latter equation to determine the curves traced out by trajectories of the system of equations for $x(t)$ and $y(t)$.

SETUP AND HYPOTHESES

Real constants λ_1, λ_2. The chain rule is valid where $x(t)$ is a differentiable function with $\dot{x} \neq 0$, so that $t \mapsto x$ is locally invertible and y can be regarded as a function of x. Here $\dot{x} = \lambda_1 x \neq 0$ requires $\lambda_1 \neq 0$ and $x \neq 0$. The cases $x = 0$ (the y-axis), $y = 0$ (the x-axis) and $\lambda_1 = 0$ are treated separately at the end.

FORMULATION

The orbit equation $\frac{dy}{dx} = \frac{\lambda_2}{\lambda_1}\frac{y}{x}$ has no explicit t and is **separable**: all the y on the left and all the x on the right.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

This is the step of the notes that turns Case 1 into a portrait. From $z_1 = C_1 e^{\lambda_1 t}$ and $z_2 = C_2 e^{\lambda_2 t}$ they eliminate t (for $C_1 > 0$):

$$z_2 = C_2\left(e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = C_2 C_1^{-\lambda_2/\lambda_1}\left(C_1 e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = \tilde{C}\, z_1^{\lambda_2/\lambda_1}.$$

The chain-rule route below reaches the same curves, $y = \tilde{C}\,|x|^{\lambda_2/\lambda_1}$ on each half-plane, without solving for t.

DERIVATION

Integration

Separate variables (for $y \neq 0$, $x \neq 0$) and integrate both sides with their antiderivatives, $\int \frac{dy}{y} = \ln|y|$:

$$\frac{dy}{y} = \frac{\lambda_2}{\lambda_1}\frac{dx}{x} \implies \ln|y| = \frac{\lambda_2}{\lambda_1}\ln|x| + C$$

Exponentiate and absorb e^C in a constant $K = \pm e^C$:

$$|y| = e^C |x|^{\lambda_2/\lambda_1} \implies \boxed{y = K\,|x|^{\lambda_2/\lambda_1}}, \qquad K \neq 0$$

Checking with the explicit solution

The equations are solved directly by $x = x_0 e^{\lambda_1 t}$, $y = y_0 e^{\lambda_2 t}$ (separation of variables, $\frac{d}{dt}(e^{-\lambda t}u) = 0$). For $x_0 \neq 0$, $|x|/|x_0| = e^{\lambda_1 t}$ so $e^t = (|x|/|x_0|)^{1/\lambda_1}$ and

$$y = y_0\left(e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = y_0\left(\frac{|x|}{|x_0|}\right)^{\lambda_2/\lambda_1} = \underbrace{y_0|x_0|^{-\lambda_2/\lambda_1}}_{K}|x|^{\lambda_2/\lambda_1}$$

which is the same family, with $K = y_0|x_0|^{-\lambda_2/\lambda_1}$ (the sign of y is the sign of y_0, preserved since $y = y_0 e^{\lambda_2 t}$).

The axes

If $x_0 = 0$ then $x(t) \equiv 0$ and the orbit is on the y-axis (with $y = y_0 e^{\lambda_2 t}$); if $y_0 = 0$ then $y \equiv 0$ (that is $K = 0$) and the orbit lies on the x-axis. The origin is the equilibrium. If $\lambda_1 = 0$, x is constant and the orbits are vertical lines.

Shapes of the curves

Exponent λ_2/λ_1	**Curves** $y = K\lvert x\rvert^r$	**Origin**
$r > 1$ (same sign, $\lvert\lambda_2\rvert > \lvert\lambda_1\rvert$)	parabola-like, tangent to the x-axis at 0	node (stable if $\lambda_i < 0$, unstable if > 0)
$r = 1$ ($\lambda_1 = \lambda_2$)	straight lines $y = Kx$ (rays)	star
$0 < r < 1$	tangent to the y-axis at 0	node
$r < 0$ (opposite signs)	hyperbola-like, $y\lvert x\rvert^{-r} = K$, asymptotic to the axes	saddle
$r = -1$	rectangular hyperbolas $xy = \pm K$	saddle

INTERPRETATION AND CHECKS

- This is the derivation behind the canonical portraits of Topic 4 Sec. 6.1: replace (x, y) by (z_1, z_2) and λ_1, λ_2 by the two eigenvalues to get $z_2 = K\lvert z_1\rvert^{\lambda_2/\lambda_1}$ (Robinson Eq. 28.12).
- Examples: $\lambda_1 = 1, \lambda_2 = 2$: $y = Kx^2$ (parabolas); $\lambda_1 = 1, \lambda_2 = -1$: $y = K/x$ (hyperbolas).
- Direction along the orbits is **not** given by the orbit equation: it is read from $\dot{x} = \lambda_1 x, \dot{y} = \lambda_2 y$ (away from 0 along an axis if $\lambda > 0$, towards it if $\lambda < 0$).
- Differentiating the answer: $\frac{dy}{dx} = K\frac{\lambda_2}{\lambda_1}\lvert x\rvert^{\lambda_2/\lambda_1 - 1}\operatorname{sign}(x) = \frac{\lambda_2}{\lambda_1}\frac{y}{x}$, which recovers the ODE (using $\lvert x\rvert^r/x = \lvert x\rvert^{r-1}\operatorname{sign} x$).